Intrapreneurship at SAP

How To Identify the most Promising Team Members and Optimally Support them

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Part I

Thesis

Abstract

Research Focus

This research concerns the inner workings of Intrapreneurship Programs. There is ample research about the antecedents that lead up to companies investing in such programs, and likewise extensive research about what happens after the end of such programs with the ideas that went through to the end, as well as those that were selected out.

To the best of our knowledge, there is no systematic, longitudinal, quantitative and qualitative research about the inner workings of such programs. In other words, previous studies have not been able to address what happens during the runtime of an Intrapreneurship, or Organizational Entrepreneurship, program. Chapter 6 on page 25 reviews the literature establishing this gap in prior research.

Against this background, this research aims to study the inner workings of intrapreneurship programs asking the following research question:

> ? How to get a winning team? How to improve finding the most promising team members that will drive their idea from submission to incubation and support them throughout the program?

from two perspectives: The program management and the participants in the program.

Chapters 1 on page 11, 3 on page 17 and 4 on page 19 describe the problem statement, as well as the aims and objectives more in detail.

Research Approach

We undertake a longitudinal study over three years following several intrapreneurship cohorts, based both on quantitative data analysis using machine learning of structured and unstructured data, as well as qualitative data analysis leveraging expert interviews and Delphi panels of members of the winning teams, as well as the program members, at the end of the program.

We are going to interact with the intrapreneurship program at three points:

1. During the stage of identification of team members, we are going to develop machine learning based classifiers that allow to identify the most promising team members.

2. During the runtime of each cohort, we are going to use machine learning based natural language processing to identify, from the content generated by the teams, aspects that indicate a winning idea.

3. At the end of each cohort, we are going to do expert interviews based on the Delphi method with both the program management as well as the members of the winning teams to validate the classifiers used at the beginning.

Chapter 7 on page 37 describes the methodological approach more in detail, and chapter **??** on page ?? discusses the expected results, as far as they can be predicted against the background of a pilot study that we executed.

Research Contribution

Because of the lack of research into the inner workings of Organizational Entrepreneurship programs, companies are investing heavily in these programs without being well informed about what works and what does not: They may focus on the ideas rather than on the teams, and when focusing on the teams, they may not be performing optimally at finding the right team members. As a result, not only could the programs perform sub-optimally, but in addition, there is a great chance of losing valuable business ideas, while giving too much time to ideas that eventually would be selected out. Chapters 2 on page 15 and 5 on page 21 deal with these topics more in detail, discussing the audience of the research, and its significance.

CHAPTER 1

Problem Statement

1.1 Research Problem

As detailed in chapter 6 on page 25, Intrapreneurship, also known as Corporate Entrepreneurship (CE) or Organisational Entrepreneurship (OE), has been studied since 1972. The topic concerns the activities of large organisations designed to harness the innovative power of startups embedded in the broader organisational context. The motivations of organisations to engage in such activities—the so-called antecedents of organisational entrepreneurship—have been the focus of numerous studies, and also the outcome of such programs.

An Organizational Entrepreneurship program typically starts with offering to the body of the organisation a platform to suggest innovative ideas for future product or service design. It then runs a selection process to select teams that, depending on the applied selection criteria, are understood to have the largest potential value for the organisation and that have the strongest indication of being able to create the traction needed to drive an idea to incubation.

The selected teams then usually undergo an acceleration process, within which a team is tasked with developing the idea to a stage where it can be either incubated, i.e., developed as part of the organization's line of products or services, or excubated, in other words spun off, as a separate business unit.

Prior research has taken different attempts to assess the readiness of organizations to engage in Corporate Entrepreneurship; in other words, the field has in extenso constructed understanding

about when and how to engage in programs that would promote innovative ideas generated from within organizations, i.e., antecedents of such programs. The more recent literature extends the research to the outcomes of these programs, including aspects of dealing with failure, and legitimacy.

What appears to be less covered in the literature is the assessment of the efficiency throughout such programs as they promote innovations inside an organization. In other words, there is a lack of research that focuses on this specific acceleration process. While there is ample research about that process in the startup community, when it comes to startups that are embedded in a larger organisational context, such research is scarce. One can only speculate why: Typically, organisations would be reluctant to offer insight into the emergence of their most innovative ideas, and even much less so in the context of a longitudinal study.

This gap in research focus has also been confirmed in a conversation the author had in 2016 with one of the leading researchers in the area, Donald Kuratko, and it entails an opportunity for contribution to practice, as Corporate Entrepreneurship programs need to justify the investment made as a function of their effectiveness to improve the quality of the teams they promote: When organisations engage in Organizational Entrepreneurship programs, they are not well informed by prior research as to how to implement the program; they not only invest substantial amounts of money into a program, they also risk turning down their most innovative workforce if they do not run the program well.

In addition, compared to external startups that only compete with the market, startups that are embedded in an organisational context also compete with the larger organisation itself: For example, they face challenges such as organisational resistance when ideas submitted to the program have the potential to compete with, and be seen as a threat by, other teams in the larger organizational context.

1.2 Research Context

The author has a strong personal motivation for this research: In 2015, the company that he is working for, SAP, one of the largest software firms globally, started an "Intrapreneurship Program" designed to inspire the development of promising new products and services of the company. The author's team won the first round of the program, which means, their idea received the green light to be funded and incubated, and then was unsuccessful in actually finding a sponsor that would host the incubation of the idea. Building on this failure, the author intends to research the what would be beneficial to a program of Corporate Entrepreneurship that strives for a successful transition of an innovative idea into an incubated corporate venture.

The research will be carried out within the organizational boundaries of SAP, where the author has been given access to the data that emerges from the ongoing Intrapreneurship program.

1.3 Research Background

As part of this research, a pilot study was undertaken between December 2017 and April 2018, attempting to validate approaches answering what was the original research question:

> ? How to measure the Quality Improvement of the Ideas, Teams, and the Program?

This research was planned to be based on an evaluation of the original idea submissions of each cohort. As part of the pilot study, 1'177 idea submissions were evaluated using a k-Nearest Neighbor based, automated text clustering mechanism. The pilot study showed that the original research question, focusing on the teams' journey and based only on the initial submissions provides for limited utility of the research: While the investigated methods proved to be relatively accurate at predicting for a given submission whether it would be promoted to a later stage or not, the predictions were heavily biased toward the earlier stages of the program, where most ideas are selected out.

Exhibit 1: Pilot Study: Predictions based on Idea Submissions

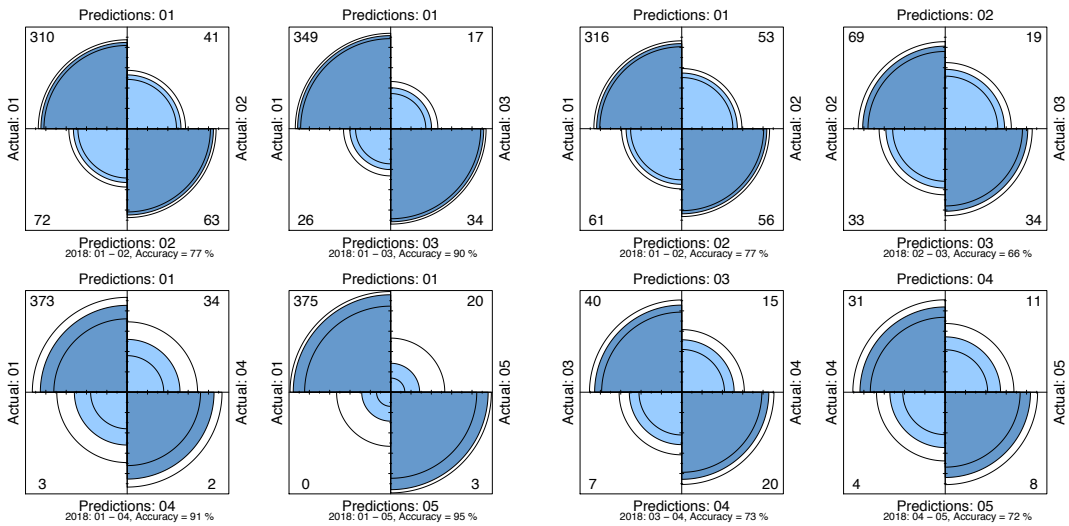
		Actual				
		Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Predictions	Phase 1	296	39	15	18	18
	Phase 2	66	35	10	9	1
	Phase 3	18	22	33	2	1
	Phase 4	0	0	1	1	1
	Phase 5	0	0	0	1	1

Total Number of Submissions	1177
Total Number of Terms	331
Total Number of Training Documents	558
Total Number of Testing Documents	558
Total Number of Correct Classifications	366
Accuracy of Prediction	62%

Figure 1.1: *Predictions based on Idea Submissions*
Source: Author.

Exhibit 1 shows the numeric outcome of the pilot study very well, and exhibit 2 illustrates the findings graphically: While the accuracy of the prediction of where an idea would end up increases as the teams move through the program—seen by the light blue circle getting smaller as the targeting gets more precise (exhibit 2 a), at the same time, the uncertainty, visualized by the radius of the white circle, increases because less and less ideas are available to train the model.

Exhibit 2: Pilot Study: Stepwise Predictions of Team Success



(a) *Stepwise Prediction: Initial Submissions* ***(b)*** *Stepwise Prediction: Iterative*

Figure 1.2: *Stepwise Predictions of Team Success*
Source: Author.

When using an alternative approach of predicting the advancement of an idea not based on its initial submission, but in addition in comparison to the stage that it survived (exhibit 2 b), the utilized machine learning approach becomes much more balanced in terms of accuracy, and the influence of uncertainty strongly decreases.

These findings of the pilot study, together with further discussions with the program management of SAP's intrapreneurship program, have prompted for a change of the research question: As the program evolves, much less emphasis is placed on the ideas themselves and much more on the selection of the right experts. This is in line with the findings of the pilot study which confirmed the utility of machine learning procedures for the identification of promising candidates from a corpus of text, but also showed the limitations of basing that analysis only on initial idea submissions.

CHAPTER 2

Audiences

There are distinct audiences to which this research intends to contribute value:

1. To Researchers that want to understand the inner workings of Intrapreneurship Programs: Attempting to close the gap in the literature that has not been able so far to look at the inner workings of intrapreneurship programs.
2. To Organisations that intend to run Intrapreneurship Programs: Providing them with reproducible methods for planning and controlling the intrapreneurship process.
3. To Innovation Teams that take part in Intrapreneurship Programs: Practical tools to improve and accelerate their decision making processes and thereby to reduce the number of pivots.

CHAPTER 3

Aims

The main aim of this research is to help understanding the very basic questions:

1. What makes intrapreneurship programs work?
2. How can the most promising team members be identified?
3. How can the efficiency of decision making within the teams be improved as they are involved in the program?

Building on these aims, and as detailed in the literature review (section 6 on page 25), this research attempts to help closing three gaps in the literature around Corporate Entrepreneurship:

3.1 Theoretical Aims

On the theoretical level, while extensive research has gone into identifying the antecedents of Corporate Entrepreneurship, its supporting organizational factors as elements of running programs, as well as the outcomes of such programs, there is a gap in the literature looking at the mechanics of such programs on the level of the identification of the teams as well as the workings of the teams as they are promoted throughout the programs.

3.2 Methodological Aims

On the methodological level, until very recently, an evaluation of potential team members of startup initiatives was mostly of qualitative nature and only flanked by quantitative evaluation of certain structured data such as revenue projections, membership networks, etc.: Most of the data is yet available as unstructured text, and text mining using machine learning approaches has developed only in recent years. The substantial investments made in intrapreneurship programs need to be justified quantitatively; hence, it appears a very valuable undertaking from a business practitioner's view to measure to which degree Corporate Entrepreneurship programs manage to improve the identification and support of the teams and their work, and machine learning techniques for natural language processing and text classification could have the potential to provide for such measures.

3.3 Practical Aims

On the practical level, we observe that a significant challenge that most startups go through is iterative, shared decision making. "Running into the wrong direction," leading to so-called pivots, is not only frustrating, it is also time-consuming and can cause the termination of an otherwise very promising idea, incurring potentially high opportunity costs. One contribution that this study intends to make by its study design is to provide for a substantial simplification of these decision making processes, using and simplifying a method called Delphi.

CHAPTER 4

Objectives

To achieve the aims set in chapter 3 on page 17, this research proposal sets three objectives for each aim that all need to be specific, measurable, attainable, realistic, and time-bound.

4.1 Theoretical Objectives

Contributing to closing the gap identified in the literature when it comes to understanding the inner workings of Corporate Entrepreneurship programs, this study wants to draw on the following three objectives:

1a Questionnaire at the Start of a Cohort: Establish Intrapreneurial, Entrepreneurial, Professional and Leadership motivation of the team members using, at the beginning of each cohort, the IEPL questionnaire (K. Y. Chan et al., 2012).

1b Questionnaire at the End of a Cohort: Establish Group Homogeneity, Goal Clarity, Cohesiveness, Process Openness and Fragmentation of the group, using, at the end of each cohort, the GPM questionnaire (Kickul, 2001). Also, establish Top Management Support, Work Discretion, Rewards and Reinforcement, Time Availability and Organizational Boundaries as characteristics of the CE program, using, at the end of each cohort, the CEAI questionnaire (Kuratko, Hornsby, and Covin, 2014).

1c Interviews at the End of a Cohort: Conduct, at the end of each cohort, expert panel interviews (simplified Delphi Method) with the team members and the program management to establish favorable and unfavorable learnings from the program.

4.2 Methodological Objectives

2a Develop an automated screening for candidates using published profiles extracted based initially on the IEPL (K. Y. Chan et al., 2012) questionnaire, and later enriched by additional learnings about favorable candidate characteristics.

2b Using Support Vector Machines (SVM) as primary, and k-Nearest Neighbors as secondary method to machine learning based text clustering (see section **??** on page ??), automate the screening for candidates.

2c Using SVM, automate the analysis of the open ended questions of the expert interviews to establish previously unknown characteristics, as well as to weigh the characteristics.

4.3 Practical Objectives

3a At the start of the research, prove the simplification to the Delphi method proposed in section **??** on page ??.

‘

3b Use this approach to achieve the qualitative rankings (section 4.1 on the previous page), and refine the method to cope with emerging issues of usability.

3c Automate this methods during the research study, writing an application that can be used in a semi-supervised manner, so as to remove the researcher from the process as much as possible.

CHAPTER 5

Significance

This chapter describes the significance of the research for its intended audiences. There are three basic contributions that this research is set out to make:

1. To Theory: Providing for a theoretical framework allowing to reduce cost and impact of errors in high- and low-efficiency programs, hence organisations to run intrapreneurship programs more effectively, therefore getting to lower total cost seen as the sum of marginal costs for introducing new ideas, and opportunity costs for missing out a relevant idea.
2. To Methodology: The ability to use machine learning to improve the screening process of team members, as well as to improve the evaluation of unstructured interview data.
3. To Practice: The ability to apply Delphi panels in a very efficient way as a means to come to agreements more quickly, and thereby to reduce the number of pivots, as well as the number of ideas that are kept in the program for too long.

It appears to this researcher that making claims about the significance at this level can tend to be vague, and therefore be unsubstantial. Therefore, this research proposal is going to show a formal model of cost estimates for the theoretical impact; it will bring forward the value of

using machine learning methods for controlling decision making process; and it will outline a practical contribution that will make the Delphi method much more easily applicable.

5.1 Theoretical Significance

A counter-argument to the relevance of this research, that an organization could bring forward, might be that their program is already very efficient at selecting teams and ideas and at improving their quality. As we show in this section, the opposite is the case.

In particular companies making this claim should be very interested in this research. To establish the significance of this research from a theoretical perspective, we are going to compare two scenarios (see Exhibit 3):

Exhibit 3: Impact of Errors

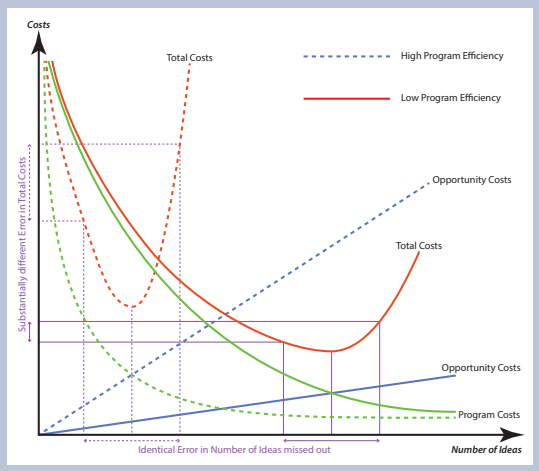


Figure 5.1: *Impact of Errors*
Source: Author.

- *High Program Efficiency:* In this scenario, the program is very efficient at selecting the teams that run through the program and in improving the quality of their ideas. Exhibit 3 shows this scenario as a set of dotted lines: Since the program is efficient at improving the quality of the ideas, the opportunity costs of missing out on one of these ideas are comparatively high; and since the program is efficient at selecting teams, the marginal cost of one idea declines strongly.

- *Low Program Efficiency:* Represented as continuous lines, in this scenario, the program is less efficient at selecting teams and at improving the quality of ideas. As a result, the marginal costs of the program decline less sharply, and likewise, because any given idea will have received less of a benefit by the program, the opportunity costs of missing out one of those ideas are lower.

This view shows that the significance of this research is higher, the higher the efficiency of the program: The impact of making an error and selecting out a team that would have had a great contribution to the organization is much higher in the high efficiency case. Therefore, particularly if an organization is already good at selecting teams and improving their ideas, this research is even so more important.

5.2 Methodological Significance

Particularly with a high number of submissions, the ranking of these teams is a tedious and error-prone process. At the same time, as established in the previous section, the more efficient the program is at selecting teams and improving the quality of their ideas, the higher the cost impact of committing errors. Therefore, using automated methods such as machine learning not only have the potential to reduce workload of the program members, but more importantly could increase the reproducibility of the decision making process. While machine learning, at its core, can only establish correlation, and never causality, the process has the potential to run in an automated way, thereby at least raising red flags when a choice made by the system is sufficiently distant from the choice made by the program members.

As the pilot study carried out as part of this research has shown, a comparatively simple, machine learning based text clustering method such as k-Nearest Neighbors can be efficiently used to separate winners from losers based purely on an analysis of unstructured text, such as idea submissions or people profiles.

5.3 Practical Significance

One aspect of the qualitative study will be to elicit the beneficial aspects of the program recurring to panels of experts in the frame of a longitudinal Delphi study. A Delphi study basically asks experts to rank characteristics as per their perceived ability to contribute to a given goal. It then collects these rankings, and calculates "Kendall's W," a factor that can be understood as a level of agreement of these experts. Next, it presents the rankings to the same panel of experts for them to reconsider their choice, and repeats the process. As a result, very typically the rankings, and hence the agreement, tend to converge; in that sense, the Delphi method can be utilized as a tool to mediate a decision-finding process.

The challenge with this approach is that the underlying mathematics—see equation (**??**) on page **??**—does not easily lend itself to the business user: Because of the complexity involved, Delphi studies are typically not executed to find agreement as part of project settings. As this study will show, by rewriting the mathematics behind Delphi, the method can be made easily accessible to the business user, and thus improve the decision making processes.

CHAPTER 6

Literature Review

6.1 Overview

This chapter gives an exhaustive review of the literature related to the research proposal: First, we describe the literature about the subject matter, i.e., Corporate Entrepreneurship (section 6.2); this allows us to establish the gaps in the literature that this research is attempting to close. Then, we give an overview of the state of the art of methodological approaches (section **??** on page ??).

6.2 Theoretical Background

From a historical perspective, this literature review contextualizes the key articles about Organizational Entrepreneurship since 1972, showing the main concepts they address and the shifts in perspectives and theory that they induce (see Exhibit 4): After first sporadic reviews of the topic (Peterson and Berger, 1971; Hill and Hlavacek, 1972) in the 1970s, the term "Intrapreneurship" is coined by Pinchot (1985), and while Burgelman (1984) theorizes that innovative initiatives inside organizations would be driven mainly from lower ranks of the organization (a claim that Hornsby, Kuratko, Shepherd, and Bott (2009) are later able to empirically disprove) the first major contradiction against Pinchot's model comes from Montago, Kuratko, and Scarcella (1986) who caution that Pinchot's model would ultimately motivate entrepreneurs to leave the organization, thereby creating a brain-drain; in other words, that rather than about structural prescripts, Corporate Entrepreneurship is about the most innovative people that an organization needs to retain. In the 1990s, Kuratko,

Montagno, and Hornsby (1990) establish their "Intrapreneurship Assessment Instrument (IAI)" quantifying three antecedents that would favor entrepreneurial engagement inside organizations.

Exhibit 4: Historical Literature Review

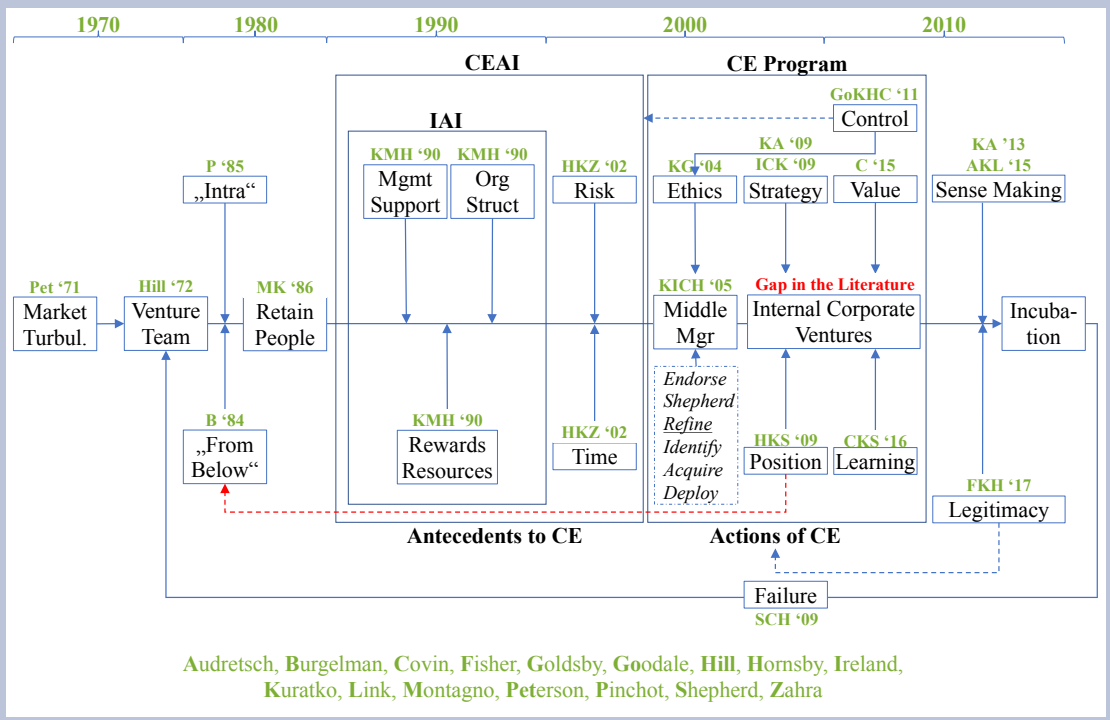


Figure 6.1: *Historical Literature Review*
Source: Author.

In 2002, Hornsby, Kuratko, and Zahra (2002) are able to extend the instrument into their "Corporate Entrepreneurship Assessment Instrument (CEAI)," establishing as five antecedents to entrepreneurial activity inside organizations: Management Support, Organizational Structure, Rewards and Resources, Risk Taking and Tolerance for Failure, as well as Time Availability. This literature, in other words, focuses mostly on what may lead to Organizational Entrepreneurship, hence its antecedents.

The literature on what happens within those programs is much more scarce: In the 2000's, Kuratko, Ireland, et al. (2005) establish the middle managers as the "lynchpins" of Corporate Entrepreneurship, and while they research what these middle managers actually *do* to further entrepreneurial behavior of organizations, Kuratko and Goldsby (2004) take an ethical

perspective and investigate what these same people should *not do*. In the early 2010s, the control question is contextualized from an organizational management angle in a highly acclaimed interdisciplinary paper by Goodale et al. (2011). Notice that with the exception of Covin, R. P. Garrett J., et al. (2015), measuring value, and Covin, Garrett, et al. (2016), focusing on learning, these articles look again at the environment within which corporate ventures would develop, rather than at the ventures themselves: In particular, Kuratko and Audretsch (2009) and Ireland, Covin, and Kuratko (2009) study Corporate Entrepreneurship activities a perspective of corporate strategy.

In recent years, Kuratko and Audretsch (2013) as well as Audretsch, Kuratko, and Link (2015) break down the different learnings and concepts about Corporate Entrepreneurship. Fisher et al. (2017) ask by whom a corporate venture team is actually legitimized as it searches for support and funding; and, in case of failure, Shepherd, Covin, and Kuratko (2009) investigate how corporate ventures can deal with the grief process in a way that these teams, who have most likely shown to be among the most valuable, value-creating assets of the corporation, can re-enter the innovation process with new ideas rather than turning away.

6.2.1 Early Research

The earliest research into the area of Organizational Entrepreneurship, according to Kuratko, Hornsby, and Hayton (2015, p. 247), "focused on the venture teams and on how entrepreneurship inside existing organization could be developed." The overall low number of articles dealt mainly with the development of new products rather than on the people driving innovative ideas (Peterson and Berger, 1971; Hill and Hlavacek, 1972, Hanan, 1976) and did not create traction in terms of research.

6.2.1.1 Market Turbulence

Linking Entrepreneurship to the strategies of companies dealing with turbulence, Peterson and Berger draw on interview data over a timespan of eight years exploring entrepreneurship as "a strategy employed by large organizations to cope with turbulent market environments." (1971, p. 91), to identify three characteristics of a successful intrapreneurship strategy:

1. As strong a segregation as possible of the entrepreneurship function from other organizational segments;
2. the linkage of entrepreneurship to a specific role that is in turn linked to a number of specialists; and
3. the minimization of financial risks of the program.

6.2.1.2 The Venture Team

Through a questionnaire study and personal interviews, Hill and Hlavacek (1972) evaluate the construct of a "venture team" as an alternative to conventional product development and marketing, as it is characterized by a single mission, relationships that are unstructured

and hence do not tend to get into the way of critical analysis, an organizational and often even geographic isolation from the parent company, and ultimately an entitlement generated by the team's venture program often being assigned to a high level within the organization. Hill and Hlavacek (1972, pp. 46–47) identify seven structural characteristics of a successful implementation of corporate entrepreneurship:

1. *Organizational separation:* Both a physical as well as a separation in reporting lines was found to be most effective.

2. *Multidisciplinarity:* Drawing venture team members not only from technical areas was seen as essential for the creativity of venture teams.

3. *Diffusion of Authority:* A flat hierarchy, independent from job descriptions and organisational level that team members otherwise had, was seen to foster an environment open to criticism of ideas.

4. *Environment of Entrepreneurship:* An arrangement of venture team members to share profits of the venture was seen as beneficial; and when team members had been asked to contribute of their own money, this "produced a high degree of involvement on the part of team members in the analysis and solution of problems." (Hill and Hlavacek, 1972, p. 47)

5. *Top Management Linkage:* The authors found the head of the venture program to typically report to a division head or C-level officer, thus establishing high-level support as well as eliminating communication problems through the hierarchy.[1]

6. *Broad Mission:* Rather than focusing on products or processes, the teams' missions tended to typically be market- and growth oriented.

7. *Flexible Life Span:* Strict deadlines and other time pressures were seen beneficial to be removed.

6.2.1.3 The Appearance of the Term Intrapreneurship"

Pinchot coins the term "intrapreneur" as a shorthand for "corporate intrapreneur" (Pinchot, 1985, p. 9) and focuses mostly on creating structural prescripts to convert large organizations, that he compares with planned economies, into networks of "intra service intraprises" (see also Pinchot, 2001), having "internal profit centers" where "service intrapreneurs" serve "internal customers" based on a "joint ownership of intraprises," "intraproperty rights," an "intracapital bank" system, etc. In other words, Pinchot, without explicitly saying so, applies concepts of transaction cost economics (Coase, 1937) to the boundaries of smaller service units that he conceives of in the framework of a larger organization.

[1] As indicated on page **??**, while this question has been brought up throughout the years, it took until last year for Fisher et al. (2017) turn the research focus to it.

6.2.1.4 How to Retain People

Montago, Kuratko, and Scarcella (1986) observe two fundamental problems with Pinchot's approach: the retention of corporate entrepreneurs, as well as the need to conceptualize the problem as mainly people driven, and to deal with it on a level of corporate strategy. In contrast to Pinchot, working with teams of each 25 middle managers across internal departmental boundaries of large organizations, the authors realize that the analysis of Corporate Entrepreneurship should begin with people, and not be departmentalized. In particular, the authors see the two issues with Pinchot's approach:

1. *Brain Drain vs. Retention of Intrapreneurs:* Pinchot appears to suggest that a true corporate entrepreneur would eventually chose to leave the company to pursue the venture he started inside, leaving only "deadwood" (Pinchot, 1985, p. 47) behind.

2. *Tame vs. Wicked Problem:* Pinchot approaches the situation as a tame problem (Grint, 2005) and hence suggests a number of out-of-the book strategies. Kuratko very early on understands that the problem is people driven that we would conceptualize and deal with it in today's terms of a wicked problem.

"In the 1980's, researchers conceptualized CE as embodying entrepreneurial behavior requiring organizational sanctions and resource commitments for the purpose of developing different types of value-creating innovations ... CE was defined simply as a process of organizational renewal" (Audretsch, Kuratko, and Link, 2015, p. 247) calling for "a significant change in the management outlook and culture." (Sathe, 1989, p. 23)

6.2.2 The Antecedents to Corporate Entrepreneurship

In the beginning of the 1990's, the research turns more formally to establish the antecedents of Corporate Entrepreneurship. Jennings and Young demand for subjective and objective measures stating that "developing both subjective and objective measures of the concept of corporate entrepreneurship would be of interest to the field of entrepreneurship." (1990, p. 53).

6.2.2.1 The Intrapreneurial Assessment Instrument (IAI)

In line with Jennings and Young, and building on their critique of Pinchot, Kuratko, Montagno, and Hornsby (1990) concentrate their research on the "managerial" part of Corporate Entrepreneurship and find that the middle managers are the key inside those organizations. As a result, they focus on perceptions, actions, outcomes and ethical consequences of corporate entrepreneurship.

The authors find that in order to understand the most effective internal environment for corporate entrepreneurial activity, an examination of antecedents to individual entrepreneurial behavior is critical. In their article "Developing an Intrapreneurial Assessment Instrument for an Effective Corporate Entrepreneurial Environment," they construct an "Intrapreneurial Assessment Instrument (IAI)." (1990, p. 49). Using Pinchot's popular word, they try to isolate

those antecedents, in other words, the most important environmental factors, of which they are able to validate the first three:

1. Management Support for Intrapreneurship
2. Organizational Structure
3. Reward and Resource Availability
4. Risk-Taking
5. Time Availability

Based on this research, Kuratko, Montagno, and Hornsby (1990) describe the development of the first instrument used to measure the environment within a corporation seeking to initiate entrepreneurial activity. This instrument, known as Intrapreneurial Assessment Instrument (IAI), containing the first three of the above-mentioned environmental factors, was then used to gather data within a large, US-based, Fortune 500 firm. Since that was a U.S. based study, the authors suggested to expand the research to larger entities, other industries, and internationally.

Conceptualizing corporate entrepreneurship as firm behavior, Zahra (1993) lays the groundwork for the following 25 years of research he did in the area of Corporate Entrepreneurship, devising a conceptual framework for firm level intrapreneurship. He reviews the validity of the Colvin-Slevin Model (Covin and Slevin, 1991) which links a company's entrepreneurial posture and its external government, strategy, internal factors and organizational performance. The author elicits revised propositions and devise a conceptual framework of firm level intrapreneurship.

Over the 10 years following Kuratko, Montagno, and Hornsby (1990), the researchers conduct a number of empirical studies to collect more data and generate working hypotheses. They also do a study outside the US, in Canada, to capture some cross-cultural effects (Hornsby, Kuratko, and Montagno, 1999).

6.2.2.2 The Corporate Entrepreneurship Assessment Instrument (CEAI)

In the Journal of Business Venturing of 2002, Hornsby, Kuratko, and Zahra (2002, p. 254) expand the earlier idea of an Interpreneurship Assessment Instrument, or IAI (Kuratko, Montagno, and Hornsby, 1990), and retitle it "Corporate Entrepreneurship Assessment Instrument (CEAI)," to get away from the Pinchot's buzzword, in favor of using the more strategic term. It is a diagnostic tool (based on perceptions) for evaluating *how supportive the corporate environment is* towards corporate venturing initiatives. With the data that the authors collected for this article, they are able to validate all five factors that they had originally hypothesized in 1990: This is hence the first publication that validated the full instrument, including the remaining two factors listed in section 5, and thereby the first publication that validated the full instrument:

1. Top Management Support
2. Work Discretion/Autonomy
3. Rewards/Reinforcement
4. Resources/Time Availability
5. Organizational Boundaries

The aforementioned emphasis on perceptions (page 29) is a relevant one: As the researchers employed the instrument, they would regularly find a different perception of the ranking of the antecedents, depending on whether they were talking to top level managers or middle level managers: A CEO would typically rank the perceived performance of his organization, as expressed by the five factors, very high, whereas when middle managers were asked (typically the authors had 5 to 6 sets of about 25 people) they would see the situation differently. Thereby, the researchers were able to show to the CEOs where they had seen perceived deficiencies:

- ▷ While a CEO might have said to have rewards at the top of the list, the middle managers might have seen it to be one of the lowest.
- ▷ While a CEO might have perceived that autonomy was given in this company for things to be done, his middle managers might be are reporting it extremely low.

As Kuratko writes, in each study they did, company after company and industry after industry, they saw this discrepancy: They had seen a perceptional problem, and therefore, developing a perceptional instrument was a powerful approach, since perception is reality: It's that person's reality. They had developed a diagnostic tool not only for researchers, but also to allow researchers to help organizations.

Building on the CEAI, Dyduch (2008) analyzes different forms of CE, with a focus on its effects of organizational performance. It contains relevant references, and in particular, its discussion of a Corporate Entrepreneurship Activity Index (Dyduch, 2008, p. 18, referring to Hornsby, Kuratko, and Zahra, 2002).

Ultimately, Kuratko, Hornsby, and Covin (2014) decided to publish the whole instrument, including the scoring tables and a description of how to use it. This is a key article, as after a decade of using the instrument, it tells *the story behind it*, and *how* and *why* to use it.

6.2.3 The Actions of Corporate Entrepreneurship

Once the antecedents of corporate entrepreneurship programs had been understood, the focus turned towards aspects of the inner workings of such programs. Yet, with very little data

available from firms about what actually happens within these programs, the research focused on factors that influence those programs.

6.2.3.1 Ethics

Kuratko and Goldsby (2004) investigate some of the ethical consequences mentioned on page 29: As managers are challenged with overcoming specific obstacles in their quest for entrepreneurial activity, the authors perceive the need to recognize where the fine line exists between clever resourcefulness and outright rule breaking, in other words, to assess which behavior differentiates the corporate entrepreneur from a rogue middle manager. They propose a framework to serve as a guideline for managers and organizations seeking to impede unethical behaviors in the pursuit of entrepreneurial activity.

6.2.3.2 Middle Managers

Because of their focus on middle managers, Kuratko, Ireland, et al. (2005) investigate the actual behavior of these middle managers in order to understand why they appear to be the *lynchpins* inside Corporate Entrepreneurship: while "the role of top-level managers revolves around the making of effective strategic decisions: Decisions that are concerned with setting the firm's direction and reaching its objectives, the role of middle-level managers focuses on effectively communicating information between the firm's two internal managerial stakeholders (top-level managers and operating-level managers)." (2005, p. 701) The authors devise a theoretical model to elicit the middle managers' behaviors for successful Corporate Entrepreneurship: Until then, while the authors had identified the *antecedents* of successful Corporate Entrepreneurship, they had not come up with a recommendation on what middle managers should actually *do*. In their article, the authors identify six behaviors that middle managers show to foster entrepreneurial behavior (2005, pp. 705–707):

- ▷ *Endorsement:* They gather innovative ideas that typically are focused on a technical level. It is their role to evaluate and ultimate endorse them.
- ▷ *Refinement*: They "convert malleable entrepreneurial opportunities into initiatives that fit the organization."
- ▷ *Shepherding*: They guide ideas and make sure they gain momentum in the organizational context.
- ▷ *Identification*: They identify resources needed for an innovation to sustain on its growth path.
- ▷ *Acquisition*: They negotiate between top level managers and idea submitters to acquire the resources that are needed, as well as people that are likely to be favorable for the implementation of a given idea.
- ▷ *Deployment*: Once acquired, middle managers ensure that resources are adequatly deployed to support the initiative.

6.2.3.3 Strategy

Kuratko and Audretsch (2009) and Ireland, Covin, and Kuratko (2009) conceptualize the components of a Corporate Entrepreneurship strategy. Building on the middle manager model, the authors develop it to a larger scale for organizations in general, at all levels, broken down by *antecedents*, *elements*, and *outcomes* of Corporate Entrepreneurship strategy:

The components of their CE strategy model include (2009):

- ▷ The *antecedents* of CE strategy, i.e., individual entrepreneurial cognitions of the organization's members and external environmental conditions that invite entrepreneurial activity;
- ▷ The *elements* of CE strategy, i.e., top management's entrepreneurial strategic vision for the firm, organizational architectures that encourage entrepreneurial processes and behavior, and the generic forms of entrepreneurial process that are reflected in entrepreneurial behavior, and
- ▷ the *outcomes* of CE strategy, i.e., organizational outcomes resulting from entrepreneurial actions, including the development of competitive capability and strategic positioning

Coakes, Smith, and Alwis (2011) propose the concept of Communities of Innovation (CoInv) in absence of formal CE programs and establish the need for these as the best foundation for sustainable innovation as well as the critical element of the CE process.

6.2.3.4 Position

In 2009, Dean Shepherd from the University of Indiana asks whether there is a difference in the way middle managers are acting in the context of Corporate Entrepreneurship, compared to top level managers above them, or first level managers below. The top level managers perceptions and behaviors had already been analyzed, but the first level managers' data had not. As a result, Hornsby, Kuratko, Shepherd, and Bott (2009) revisit the data and ask the question: With regards to the antecedents elicited in the CEAI, are they supported from all organizational levels? In particular, they look at the number of ideas implemented, not just suggested, and find an interesting relationship with the level of the organization that an idea would originate from: If an idea had originated from a senior level, it was likely that the idea was going to be implemented; for middle managers, some ideas were implemented. But for first line managers, *not a single idea was implemented*, hence empirically invalidating prior theory of Burgelman (1983).

6.2.3.5 Value

Covin, R. P. Garrett J., et al. (2015) build and test a model of venture performance drawing on data from 145 internal corporate ventures (ICVs). They find that the evolution of value proposition is related to venture performance in a curvilinear manner. Specifically, the article establishes two instruments, one for managers of the ventures, and one for the parental, senior

managers over them, and finds that ICVs whose propositions exhibit moderate evolution perform better than those that show either no or extensive evolution; in addition, the value proposition evolution-performance relationship is moderated by the degree to which the parent corporation is familiar with the venture's target market.

6.2.3.6 Control

Building on the prior work on ethical consequences Kuratko and Goldsby (2004), Goodale et al. (2011) examine control factors as the field started arguing that to have entrepreneurial activity, one needed to release all controls. In their empirical and highly acclaimed article, they study 177 firms operating on a large variety of industries to investigate the effect of several commonly-acknowledged antecedents of Corporate Entrepreneurship, as measured by the CEAI, on innovation performance. They then examined the moderating effects of operations control variables, specifically risk control and process control formality, on the relationships between the antecedents of Corporate Entrepreneurship and innovation performance. The study finds that each of the five antecedents significantly interacts with one or both of the operations control variables and thus has an influence on innovation performance.

Published in the Journal of Operations Management and bridging the gap between Operations Management and Entrepreneurship, this article has had a large impact and has been awarded the 2016 Ambassador Award as it was recognized that it has not only been widely cited in the subsequent literature, but also that a significant portion of these citations have appeared in non-Operations Management journals.

6.2.3.7 Learning

While ICV learning proficiency appears to be differentially related to venture performance, depending on how various aspects of business planning are approached and evolve Covin, R. P. Garrett J., et al. (2015), Covin, Garrett, et al. (2016) find that ICV learning proficiency appears to be more positively related to venture performance in cases where the ICVs' initial value propositions are unclear, and when their goals do not extensively evolve over the course of the ventures' development. Specifically, this research looks at the parent company and its controls one the one hand, and knowledge base on the other, to examine the degree to which ICVs are either more autonomous in their development and learning, or under stricter control by their parent.

Sambrook and Roberts (2005) give an extensive literature review on different models of Corporate Entrepreneurship in the context of and Organisational Learning. No analysis is made of the actual inner workings of Corporate Entrepreneurship programs, but their work is useful to understand the influencers of Corporate Entrepreneurship.

Zahra (2015) asks how CE creates knowledge, what knowledge does it create, and what is the role of entrepreneurial hubs—such as formal CE programs—in capturing, accumulating, translating and integrating that knowledge.

6.2.3.8 Motivation

Turner and Pennington (2015) develop a Motivation, Opportunity, and Ability framework (MOA) which can potentially be used as a prescriptive and evaluation tool for the evaluation of Corporate Entrepreneurship, and that can potentially act as a prescriptive tool for the evaluation of Corporate Entrepreneurship where firms and researches alike can utilize a MOA analysis to evaluate corporate entrepreneurship efforts within an organization. The framework supplements traditional Corporate Entrepreneurship measures such as the Corporate Entrepreneurship Assessment Instrument (Kuratko, Montagno, and Hornsby, 1990; Hornsby, Kuratko, and Zahra, 2002).

K.-Y. Chan et al. (2017) develop a self-report measure of employee intrapreneurial motivation based on the recently proposed Entrepreneurship, Professionalism and Leadership (EPL) framework (K. Y. Chan et al., 2012) based on a 27 items measurement scale, as well as an Intrapreneurial Motivation Scale comprised of 9 items, to address the question, "who wants to be an intrapreneur?" (K.-Y. Chan et al., 2017, p. 3) Their study, together with the CEAI (Kuratko, Hornsby, and Covin, 2014), provides for an analytical framework that this research will expand on.

6.2.3.9 Team Performance

Using Hackman's (1987) Input-Process-Output (IPO) model, Kickul (2001) investigates the theoretical relationship between individual personality factors and group performance that are mediated by five group variables:

1. Group Homogeneity
2. Group Goal Clarity
3. Group Cohesiveness
4. Open Group Process
5. Internal Fragmentation

CHAPTER 7

Methodology

7.1 Overview

Blaikie (2010, p. 81) gives an analytical prescript starting with the research strategy based on the kind of research question asked, from which logically follow a choice of ontology, epistemology, methodology, and, ultimately, methods. Easterby-Smith, Thorpe, and Jackson (2015, p. 47) visualize these as layers of a tree, with ontology as its core (see Exhibit 5).

Exhibit 5: The Four Rings Model

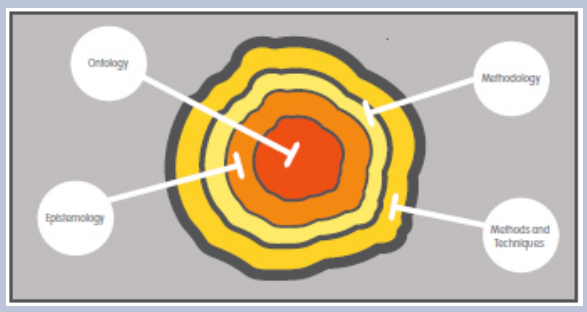


Figure 7.1: *The Four Rings Model*
Source: Easterby-Smith, Thorpe, and Jackson, 2015, p. 47.

7.2 Research Strategy

As introduced on page 9, the research question is a how question about a change process, which, according to Blaikie (2010, p. 105) would best be addressed using a retroductive research strategy. This strategy aims "to discover underlying mechanisms that, in particular contexts, explain observed regularities." (Blaikie, 2010, p. 87) The first stage, according to Blaikie, is "to provide an

adequate description of the regularity to be explained" (Blaikie, 2010, ibd.), using, in this case, an retroductive strategy that includes an examination of the context, the mechanisms, and their relevance, or weights, to establish the "features of the context, in terms of ways in which it facilitates or inhibits the operation of the mechanism(s)." (Blaikie, 2010, ibd.) Blaikie indicates that "the central problem for the retroductive research strategy is how to discover the structures and mechanisms that are proposed to explain observed regularities." (Blaikie, 2010, ibd.)

As we shall show in the following sections, we plan to employ a formal, retroductive detection of these features using a data mining approach. Suffice it to say at this point that the language used by Blaikie is very congruent with the formulations of current machine learning methods that attempt to retroductively construct theory from data, in other words, that "let the data speak" (Strauss, 1987, p. 217).

7.3 Research Philosophy

This author takes the stance of a *Depth Realist*. This means that compared to a shallow realist stance, we go yet a bit further in actually formally defining structural layers of reality: what can be empirically observed, what actually exists, and what really happens but cannot be readily observed. This gives depth to reality (hence the name), and allows to extend the view to dynamic processes. On the epistemological side, this corresponds to a Neo-Realist stance, where the knowledge of the underlying causes for the observable mechanisms are derived from the mechanisms that produce them (and which cannot be observed). This knowledge is formulated by conceptualizing underlying entities, or mechanisms, which can now be congruent with one another, but which can also compete; while this appears to be complicated, it becomes very clear thinking about, for example, a multilinear regression: Each of its factors can have his own influence on a given outcome, and the direction of that influence on the level of the outcome can be positive or negative. With more than one independent variables, these influences can either reinforce one another, or they can cancel out. It is by measuring these influences and generating models about them that we create knowledge about the underlying structure of reality and hence make proposals for, or assumptions of, what really happens. Without going too much into detail, correspondingly, in Machine Learning terminology, the layered approach of the Depth-Realist ontology is very much congruent with the concepts of hidden layers in a neural network, a kernel function in the context of support vector machines, or more generally the concept of a model in machine learning.

7.3.1 Research Question

It is important to recall the research question at this point:

> ? How to get a winning team? How to improve finding the most promising team members that will drive their idea from submission to incubation and support them throughout the program?

7.3.2 Units of Analysis

7.3.2.1 The Individual

The research question calls for a method to identify the most promising candidates, i.e., it's first unit of analysis is the candidate herself. At this level, we intend to use K.-Y. Chan et al.'s (2017) Intrapreneurial, Entrepreneurial, Professional, and Leadership (IEPL) motivation scale to identify promising team members and assess their personality.

7.3.2.2 The Team

The candidates form a winning team, so that their personalities are mediated by the inner workings of the team. The unit of analysis here becomes the team, and we intend to use Kickul's [a#Kickul:2001aa]'s (2001) Group-Performance-Model (GPM) to assess the mediating effect of the characteristics of Group Homogeneity, Group Goal Clarity, Group Cohesiveness, Open Group Process and Internal Fragmentation.

7.3.2.3 The Program

The groups are supported by the intrapreneurship program. On this level, the unit of analysis is therefore the program itself, and we intend to use Kuratko, Hornsby, and Covin's (2014) Corporate Entrepreneurship Assessment Instrument (CEAI) to assess the mediating effect of the program.

7.3.3 Measures

7.3.3.1 Individual Level

On the individual level, there are four measures to assess the individual's propensity to act as an intrapreneur (K.-Y. Chan et al., 2017):

- ▷ *Intrapreneurial Motivation:* Reflecting the individual's interest and motivation to take part in intrapreneurial and innovative engagements within their organization
- ▷ *Entrepreneurial Motivation:* Aspiration of the individual's readiness to value, and engage in, entrepreneurial activities outside their organization
- ▷ *Professional Motivation:* Reflecting the individual's motivation to, and valuation of, professional advancement
- ▷ *Leadership Motivation:* Assessing the individual's aspiration to take a leadership position

7.3.3.2 Team Level

On the level of the team, there are five measures to assess the team performance (Cammann et al., 1983; Kickul, 2001)

- ▷ *Group Homogeneity:* Measures the similarity of the group members with regards to characteristics such as background and experience
- ▷ *Group Goal Clarity:* The understanding that the team member has of the group's goals

- ▷ *Group Cohesiveness:* Measures the social characteristics of the group and how attractive the membership in a group is to people outside the group
- ▷ *Open Group Process:* Measures the level of satisfaction a team member gets from his involvement, which results in a reduction of conflicts
- ▷ *Internal Fragmentation:* Measures the degree of group conflict as well as the interpersonal relationship within the group

7.3.3.3 Program Level

On the program level, there are five measures to assess the antecedents of Organizational Entrepreneurship (Kuratko, Hornsby, and Covin, 2014):

- ▷ *Top Management Support:* The extent to which top managers are perceived to support and promote entrepreneurial behavior
- ▷ *Work Discretion:* The extent to which the organization is perceived to tolerate failure and delegate authority and responsibility to lower levels
- ▷ *Rewards and Reinforcement:* The extent to which the organization is perceived to provide for systems rewarding entrepreneurial activity and encouraging risk taking and innovation
- ▷ *Time Availability:* The individual's perception of their work schedule allowing for extra time to pursue innovative activities
- ▷ *Organizational Boundaries:* The extend to which the boundaries within an organization are perceived to be flexible, allowing for a flow of information within the organization, as well as across the boundaries of the firm (Coase, 1937)

7.3.4 Hypotheses

In the theoretical framework (see Exhibit **??** on page ??), every team member brings into the group, and hence the program, his own personality and experience. Acting together as one team, the group must perform successfuly to be selected by the program. The group dynamic thus mediates the individual performances of the team members, and likewise, as the group is embedded within the organization's intrapreneurship program, the characteristics of that organization mediate the final outcome.

While the first part of the research question deals with operationalizing the identification of the most promising team members, the second part concerns the program's support to drive their idea to success. Within that process, what is the relationship between the individual personality, the group characteristics, and those of the organization? This section looks at the influencing factors derived from the literature and draws hypotheses.

7.3.4.1 Individual Level

The variables Intrapreneurial Motivation, Entrepreneurial Motivation, Professional Motivation and Leadership Motivation are all positively related to a team's performance through the mediating group-process and program variables.

7.3.4.2 Team Level

The variables Group Goal Clarity, Group Cohesiveness and Open Group Process are all positively, the variables Group Homogeneity and Internal Fragmentation negatively related to a team's performance through the mediating program variables.

7.3.4.3 Program Level

The variables Top Management Support, Work Discretion, Rewards and Reinforcement, Time Availability and Organizational Boundaries are all positively reinforcing characteristics related to the teams' performance.

Notes

References

Audretsch, D. B., Kuratko, D. F., & Link, A. (2015). 'Making sense of the elusive paradigm of entrepreneurship.', *Small Business Economics*, 45 (4), 703–712.
(Cit. on pp. 27, 29.).

Blaikie, N. (2010). *Designing Social Research* (2nd ed.). Polity Press.
(Cit. on pp. 37, 38.).

Burgelman, R. A. (1983). 'A process model of internal corporate venturing in the diversified major firm.', *Administrative Science Quarterly*, 28 (2), 223–244.
(Cit. on p. 33.).

Burgelman, R. A. (1984). 'Designs for corporate entrepreneurship in established firms.', *California Management Review*, 26 (3), 154–166.
(Cit. on p. 25.).

Cammann, C., Fichman, M., Jenkins, D. G., & Klesh, J. R. (1983). *Assessing the Attitudes and Perceptions of Organizational Members: Assessing Organizational Change*. Wiley-Interscience.
(Cit. on p. 39.).

Chan, K. Y., Ho, M.-h. R., Chernyshenko, O. S., Bedford, O., Uy, M. A., Gomulya, D., Sam, Y., & Phan, W. M. J. (2012). 'Entrepreneurship, Professionalism, Leadership: A Framework and Measure for understanding Boundaryless Careers', *Journal of Vocational Behavior*, 81, 73–88.
(Cit. on pp. 19, 20, 35.).

Chan, K.-Y., Ho, M.-H. R., Kennedy, J. C., Uy, M. A., Kang, B. N. Y., Chernyshenko, O. S., & Zu, K. Y. T. (2017). 'Who Wants to Be an Intrapreneur? Relations between Employees' Entrepreneurial, Professional,and Leadership Career Motivations and Intrapreneurial Motivation in Organizations', *Frontiers in Psychology*, 8 (1), 1–11.
(Cit. on pp. 35, 39.).

Coakes, E. W., Smith, P. A. C., & Alwis, D. (2011). 'Sustainable Innovation and Right to Market', *Information Systems Management*, 28 (1), 30–42.
(Cit. on p. 33.).

Coase, R. H. (1937). The Nature of the Firm. In Williamson, Oliver E. / Winter, Sydney G. (1991) (Ed.), *The Nature of the Firm: Origins, Evolution and Development* (pp. 18–33). Oxford University Press.
(Cit. on pp. 28, 40.).

Covin, J. G., & Slevin, D. P. (1991). 'A Conceptual Model of Entrepreneurship as Firm Behavior', *Entrepreneurship: Theory & Practice*, 16 (1), 7–25.
(Cit. on p. 30.).

Covin, J. G., Garrett, R. P., Jr., Kuratko, D. F., & Shepherd, D. A. (2015). 'Value proposition evolution and the performance of internal corporate ventures', *Journal of Business Venturing*, 30, 749–774.
(Cit. on pp. 27, 33, 34.).

Covin, J. G., Garrett, R. P., Gupta, J. P., Kuratko, D. F., & Shepherd, D. A. (2016). 'The Interdependence of Planning and Learning Among Internal Corporate Ventures', *Entrepreneurship: Theory & Practice*, 1–34. https://doi.org/10.1111/etap.12265
(cit. on pp. 27, 34.).

Dyduch, W. (2008). 'Corporate Entrepreneurship Measurement for improving Organizational Performance', *Journal of Economics & Management*, 4, 15–40.
(Cit. on p. 31.).

Easterby-Smith, M., Thorpe, R., & Jackson, P. R. (2015). *Management & Business Research* (5th ed.). Sage Publications.
(Cit. on p. 37.).

Fisher, G., Kuratko, D. F., Bloodgood, J. M., & Hornsby, J. S. (2017). 'Legitimate to whom? The challenge of audience diversity and new venture legitimacy', *Journal of Business Venturing*, 32, 52–71.
(Cit. on pp. 27, 28.).

Goodale, J. C., Kuratko, D. F., Hornsby, J. S., & Covin, J. G. (2011). 'Operations management and corporate entrepreneurship: The moderating effect of operations control on the antecedents of corporate entrepreneurial activity in relation to innovation performance', *Journal of Operations Management*, 29 (Special Issue Operations management, entrepreneurship, and value creation: Emerging opportunities in a cross-disciplinary context), 116–127.
(Cit. on pp. 27, 34.).

Grint, K. (2005). 'Problems, problems, problems: The social construction of leadership', *Human Relations*, 58 (11), 1467–1494.
(Cit. on p. 29.).

Hackman, J. R. (1987). The Design of Work Teams. In J. W. Lorsch (Ed.). Prentice Hall.
(Cit. on p. 35.).

Hanan, M. (1976). 'Venturing corporations–think small to stay strong.', *Harvard Business Review*, 54 (3), 139–148.
(Cit. on p. 27.).

Hill, R. M., & Hlavacek, J. D. (1972). 'The Venture Team: A New Concept in Marketing Organization', *Journal of Marketing*, 36 (3), 44–50.
(Cit. on pp. 25, 27, 28.).

Hornsby, J. S., Kuratko, D. F., & Montagno, R. V. (1999). 'Perception of internal factors for corporate entrepreneurship: A comparison of canadian and u.s. managers', *Entrepreneurship: Theory & Practice*, 24 (2), 11–26.
(Cit. on p. 30.).

Hornsby, J. S., Kuratko, D. F., Shepherd, D. A., & Bott, J. P. (2009). 'Managers' corporate entrepreneurial actions: Examining perception and position', *Journal of Business Venturing*, 24 (Special Issue Corporate Entrepreneurship), 236–247.
(Cit. on pp. 25, 33.).

Hornsby, J. S., Kuratko, D. F., & Zahra, S. A. (2002). 'Middle managers' perception of the Internal Environment for Corporate Entrepreneurship: Assessing a Measurement Scale', *Journal of Business Venturing*, 17, 253–273.
(Cit. on pp. 26, 30, 31, 35.).

Ireland, R. D., Covin, J. G., & Kuratko, D. F. (2009). 'Conceptualizing corporate entrepreneurship strategy', *Entrepreneurship: Theory & Practice*, 33 (1), 19–46.
(Cit. on pp. 27, 33.).

Jennings, D. F., & Young, D. M. (1990). 'An empirical comparison between objective and subjective measures of the product innovation domain of corporate entrepreneurship.', *Entrepreneurship: Theory & Practice*, 15 (1), 53–66.
(Cit. on p. 29.).

Kickul, G. (2001). 'Antecedents Of Work Team Performance in a Business Simulation: Personality and Group Interaction', *Developments in Business Simulation and Experiential Learning*, 28, 128–136.
(Cit. on pp. 19, 35, 39.).

Kuratko, D. F., & Audretsch, D. B. (2009). 'Strategic entrepreneurship: Exploring different perspectives of an emerging concept.', *Entrepreneurship: Theory & Practice*, 33 (1), 1–17.
(Cit. on pp. 27, 33.).

Kuratko, D. F., & Audretsch, D. B. (2013). 'Clarifying the domains of corporate entrepreneurship.', *International Entrepreneurship And Management Journal*, 9 (3), 323–335.
(Cit. on p. 27.).

Kuratko, D. F., & Goldsby, M. G. (2004). 'Corporate entrepreneurs or rogue middle managers? a framework for ethical corporate entrepreneurship', *Journal of Business Ethics*, (1), 13.
(Cit. on pp. 26, 32, 34.).

Kuratko, D. F., Hornsby, J. S., & Covin, J. G. (2014). 'Diagnosing a firm's internal environment for corporate entrepreneurship', *Business Horizons*, 57, 37–47.
(Cit. on pp. 19, 31, 35, 39, 40.).

Kuratko, D. F., Hornsby, J. S., & Hayton, J. (2015). 'Corporate entrepreneurship: the innovative challenge for a new global economic reality', *Small Business Economics*, 45 (2), 245–253.
(Cit. on p. 27.).

Kuratko, D. F., Ireland, R. D., Covin, J. G., & Hornsby, J. S. (2005). 'A model of middle-level managers' entrepreneurial behavior', *Entrepreneurship: Theory & Practice*, 29 (6), 699–716.
(Cit. on pp. 26, 32.).

Kuratko, D. F., Montagno, R. V., & Hornsby, J. S. (1990). 'Developing an intrapreneurial assessment instrument for an effective corporate entrepreneurial environment', *Strategic Management Journal*, 11 (4), 49–58.
(Cit. on pp. 25, 29, 30, 35.).

Montago, R. V., Kuratko, D. F., & Scarcella, J. H. (1986). 'Perception of entrepreneurial success characteristics .', *American Journal of Small Business*, 10 (3), 25–32.
(Cit. on pp. 25, 29.).

Peterson, R. A., & Berger, D. G. (1971). 'Entrepreneurship in organizations: Evidence from the popular music industry', *Administrative Science Quarterly*, 16 (1), 97–106.
(Cit. on pp. 25, 27.).

Pinchot, G. (1985). *Intrapreneuring: Why You Don't Have to Leave the Corporation to Become an Entrepreneur*. Harper & Row.
(Cit. on pp. 25, 28, 29.).

Pinchot, G. (2001). 'Free Intraprise', *Executive Excellence*, 18 (1), 10.
(Cit. on p. 28.).

Sambrook, S., & Roberts, C. (2005). 'Corporate Entrepreneurship and Organizational Learning: A Review of the Literature and the Development of a Conceptual Framework', *Strategic Change*, 14 (3), 141–155.
(Cit. on p. 34.).

Sathe, V. (1989). 'Fostering entrepreneurship in the large, diversified firm.', *Organizational Dynamics*, 18 (1), 20.
(Cit. on p. 29.).

Shepherd, D. A., Covin, J. G., & Kuratko, D. F. (2009). 'Project failure from corporate entrepreneurship: Managing the grief process', *Journal of Business Venturing*, 24, 588–600.
(Cit. on p. 27.).

Strauss, A. L. (1987). *Qualitative Analysis for Social Scientists*. Cambridge University Press. (Cit. on p. 38.).

Turner, T., & Pennington, W. (2015). 'Organizational Networks and the Process of Corporate Entrepreneurship: How the Motivation, Opportunity, and Ability to Act affect Firm Knowledge, Learning, and Innovation', *Small Business Economics*, 45 (2), 447–463. (Cit. on p. 35.).

Zahra, S. A. (1993). 'A Conceptual Model of Entrepreneurship as Firm Behavior: A Critique and Extension', *Entrepreneurship: Theory & Practice*, 17 (4), 5–21. (Cit. on p. 30.).

Zahra, S. A. (2015). 'Corporate Entrepreneurship as Knowledge Creation and Conversion: The Role of Entrepreneurial Hubs', *Small Business Economics*, 44 (4), 727–735. (Cit. on p. 34.).